

CS 302.1 - Automata Theory

Lecture 03

Shantanav Chakraborty

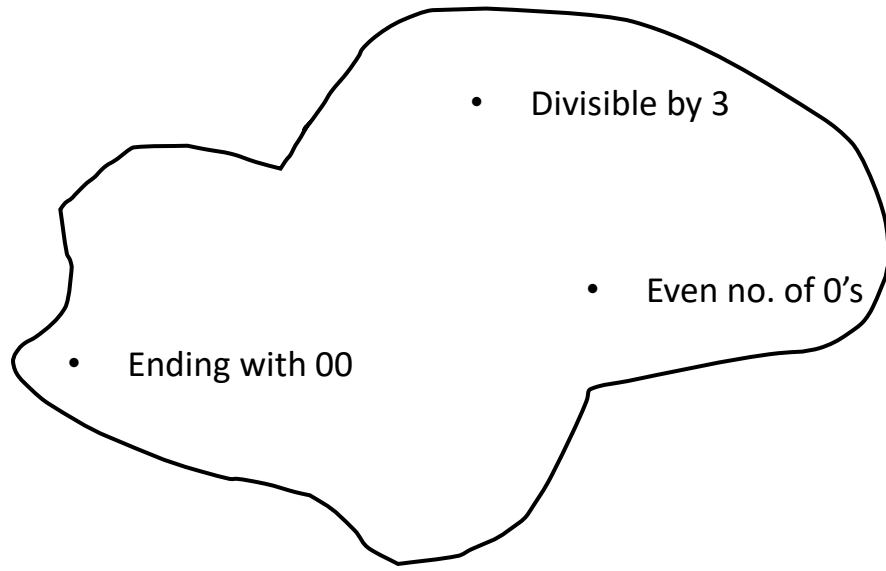
Center for Quantum Science and Technology (CQST)

Center for Security, Theory and Algorithms (CSTAR)

IIIT Hyderabad



Quick Recap



A language is called a **Regular Language** if there exists some finite automata deciding it.

If M be a finite automaton (DFA/NFA) and,

$$L(M) = \{\omega | \omega \text{ is accepted by } M\}$$

$L(M)$ is regular.

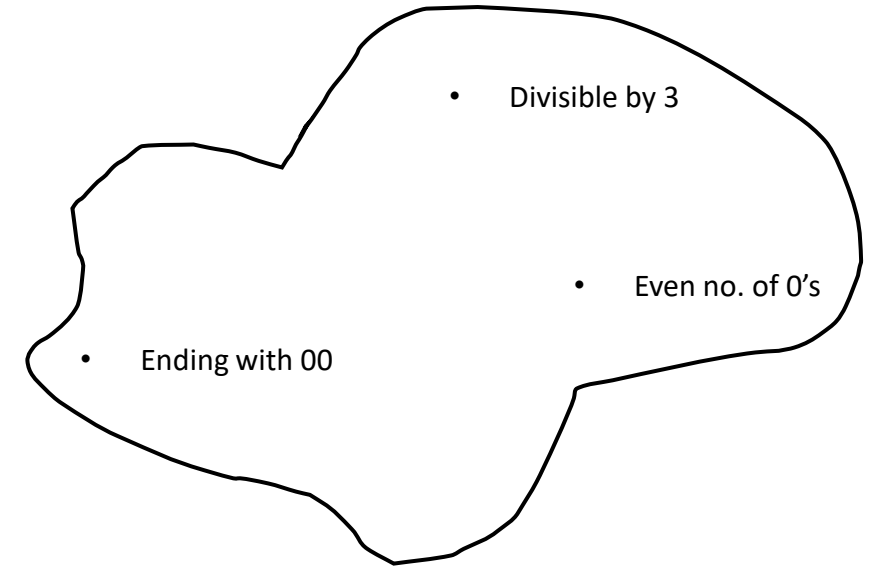
- DFAs and NFAs are equivalent
- For every NFA we can obtain a “Remembering DFA” that accepts the same language.

Quick Recap

Regular Operations:

Let L_1 and L_2 be languages. The following are the *regular operations*:

- **Union:** $L_1 \cup L_2 = \{x | x \in L_1 \text{ or } x \in L_2\}$
- **Concatenation:** $L_1 \cdot L_2 = \{xy | x \in L_1 \text{ and } y \in L_2\}$
- **Star:** $L_1^* = \{x_1 x_2 \cdots x_k | k \geq 0 \text{ and each } x_i \in L_1\}$



Set of all regular Languages

Star operation: It is a unary operation (unlike the other two) and involves putting together *any number of strings in L_1 together to obtain a new string.*

Note: Any number of strings includes “0” as a possibility and so the empty string ϵ is a member of L_1^* .

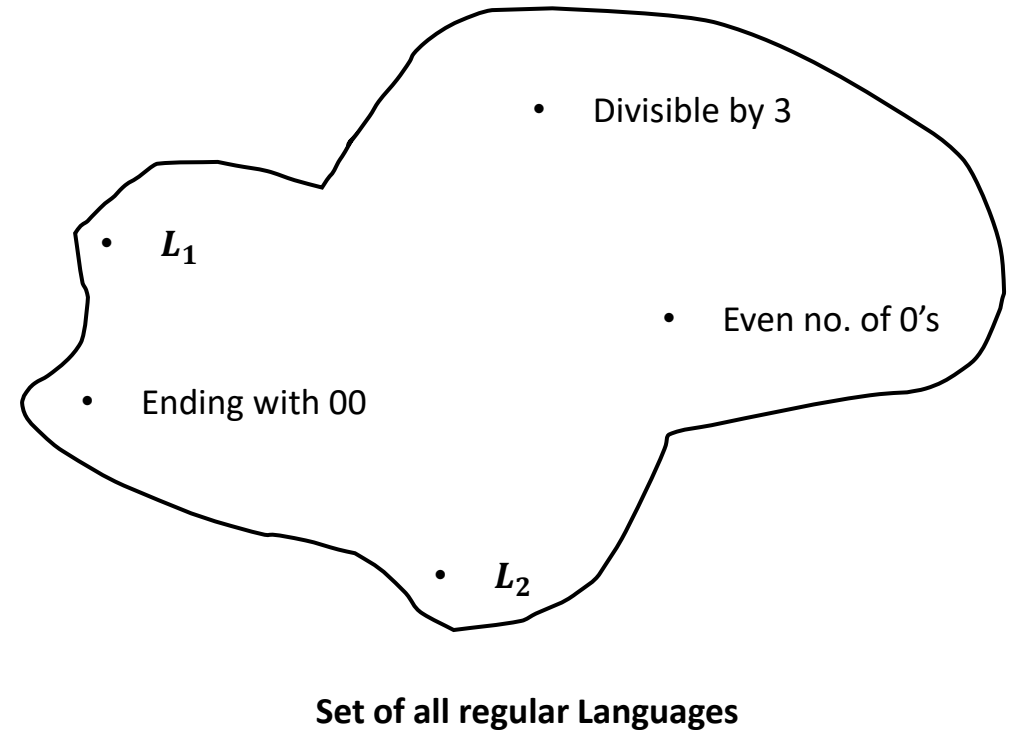
$$\text{If } \Sigma = \{a\}, \Sigma^* = \{\epsilon, a, aa, aaa, \dots\}; \text{ If } \Sigma = \{\Phi\}, \Sigma^* = \{\epsilon\}$$

Quick Recap

We want to check whether the set of regular languages are **closed** under some operations.

What does this mean?

- We pick up points within the set of all regular languages (say L_1 and L_2)
- Perform *set operations* such as Union, concatenation, Star, intersection, complement etc on them.
- Observe whether the resulting language still belongs to the set of all regular languages.
- If so, we say, regular languages are **closed** under that operation.

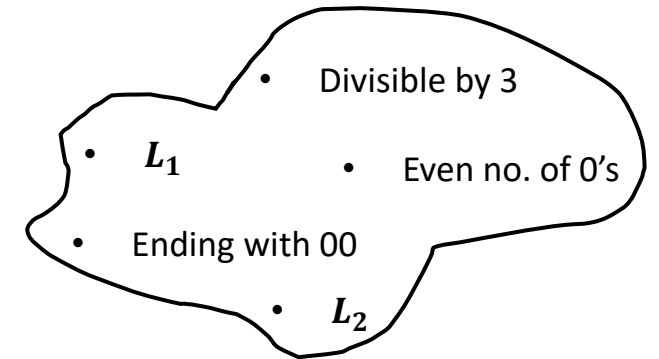


For example, the **natural numbers** are **closed under addition/multiplication** and **not under subtraction/division**.

Closure of Regular Languages

Q: Is the set of all regular languages **closed under union**?

Suppose L_1 and L_2 are regular languages. Is $L = L_1 \cup L_2$ also regular?



Set of all regular Languages

Closure of Regular Languages

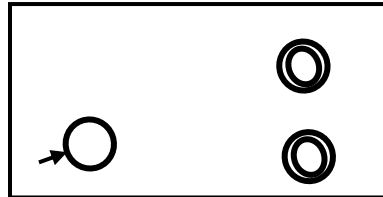
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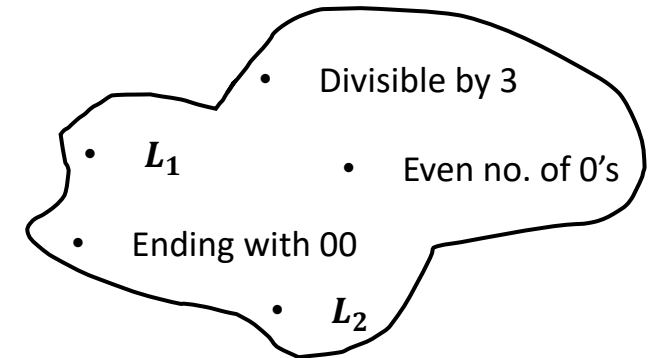
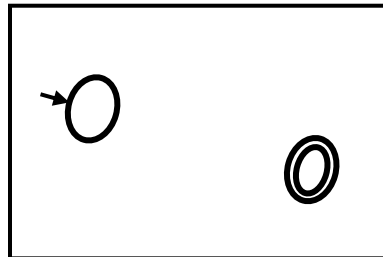
Proof: Since L_1 and L_2 are regular, there must be a DFA M_1 that accepts L_1 , i.e. $L(M_1) = L_1$ and a DFA M_2 that accepts L_2 , i.e. $L(M_2) = L_2$.

Using M_1 and M_2 , we will show how to construct an NFA M that accepts $L = L_1 \cup L_2$, i.e. $L(M) = L_1 \cup L_2$.

Suppose the DFA M_1 is



And the DFA M_2 is



Set of all regular Languages

Closure of Regular Languages

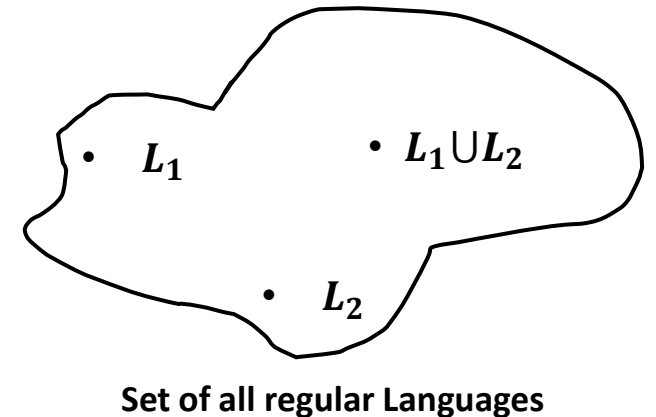
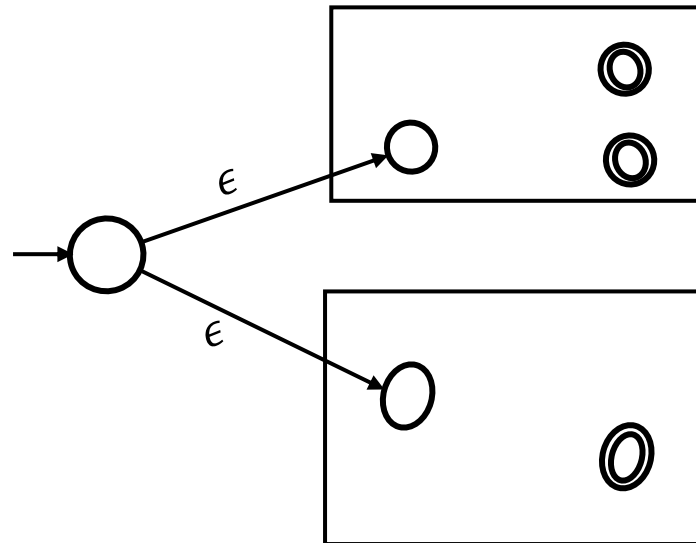
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NFA M accepting $L = L_1 \cup L_2$



Closure of Regular Languages

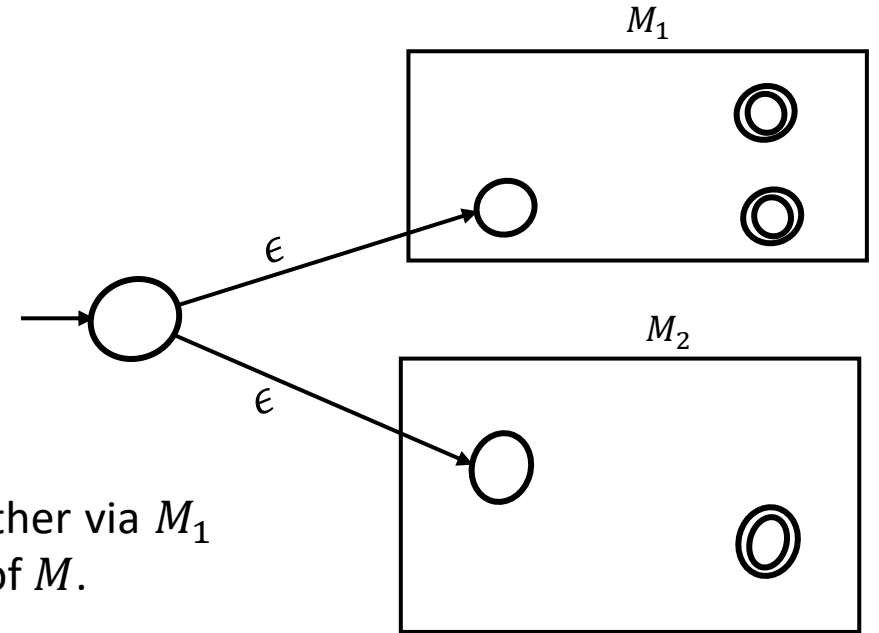
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Suppose L_1 and L_2 are regular languages. Is $L = L_1 \cup L_2$ also regular?

Proof: In order to prove that $L(M) = L_1 \cup L_2$, we show two things:

(i) $L \subseteq L_1 \cup L_2$

Let $\omega \in L$, i.e. ω is accepted by M . The final state for L can be reached either via M_1 or M_2 . Thus ω must be accepted by either of them to reach the final state of M .



Closure of Regular Languages

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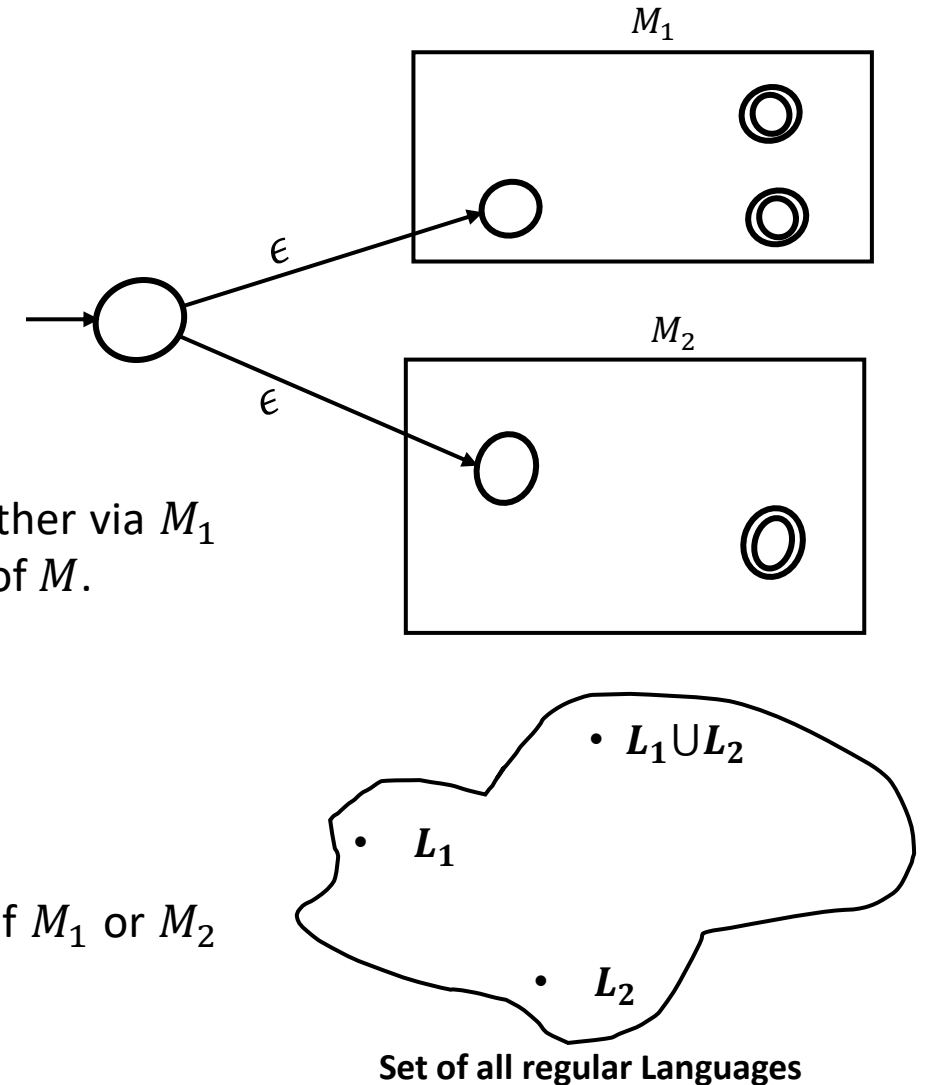
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(ii) $L_1 \cup L_2 \subseteq L$

Let $\omega \in L_1 \cup L_2$. Then, $\omega \in L_1$ or $\omega \in L_2$.

Thus, ω must reach the final state of M_1 or M_2 . But since the start state of M_1 or M_2 can be reached from the start state of M by taking an ϵ -transition, $\omega \in L$.

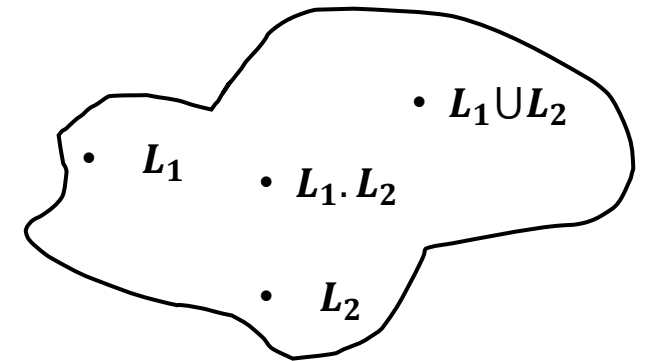
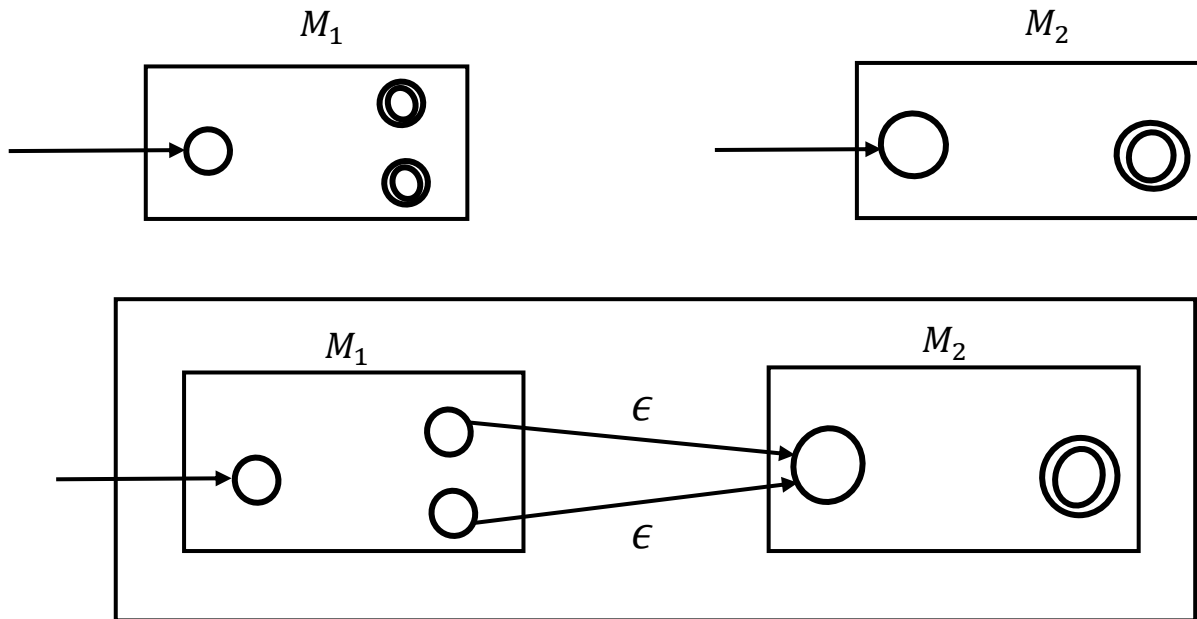


Closure of Regular Languages

Q: Is the set of all regular languages **closed under concatenation**? Suppose L_1 and L_2 are regular languages. Is $L = L_1.L_2$ also regular?

Proof: Since L_1 and L_2 are regular, there must be a DFA M_1 that accepts L_1 , i.e. $L(M_1) = L_1$ and a DFA M_2 that accepts L_2 , i.e. $L(M_2) = L_2$.

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Set of all regular Languages

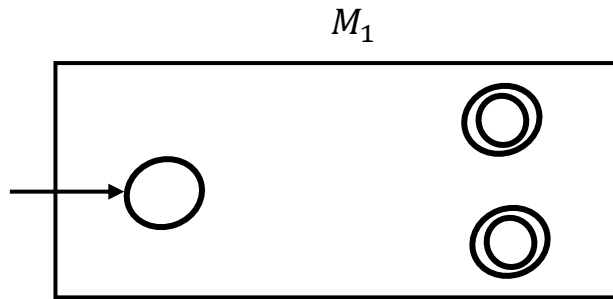
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NFA M accepting $L = L_1.L_2$

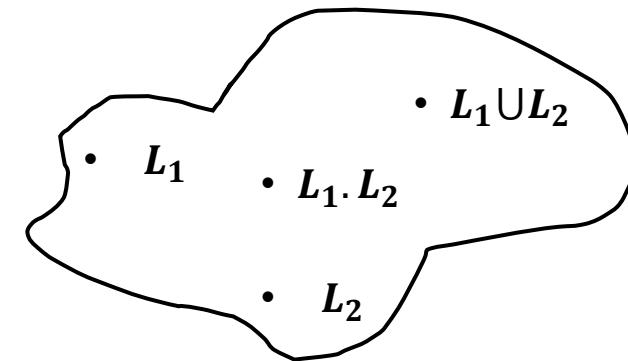
Closure of Regular Languages

Q: Is the set of all regular languages **closed under star**? Suppose L_1 is a regular language. Is L_1^* also regular?

Proof: Since L_1 is regular, there must be a DFA M_1 that accepts L_1 , i.e. $L(M_1) = L_1$. Using M_1 , we will show how to construct an NFA M that accepts $L = L_1^*$.



$$L_1^* = \{x_1x_2 \cdots x_k \mid k \geq 0 \text{ and each } x_i \in L_1\}$$

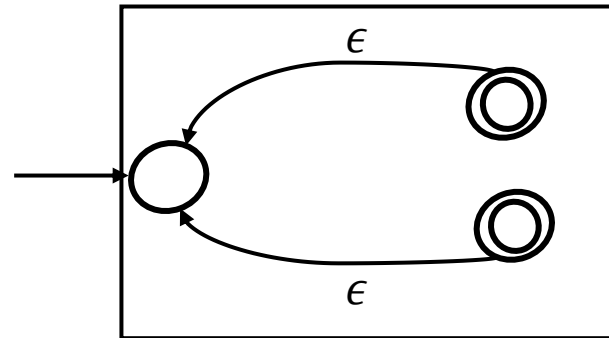
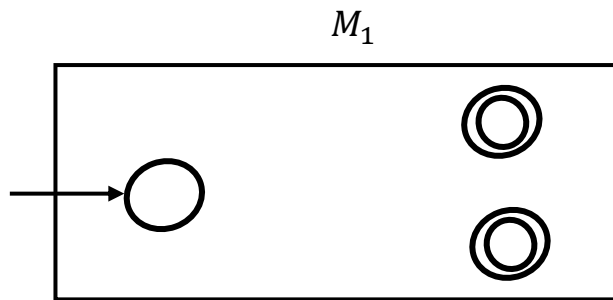


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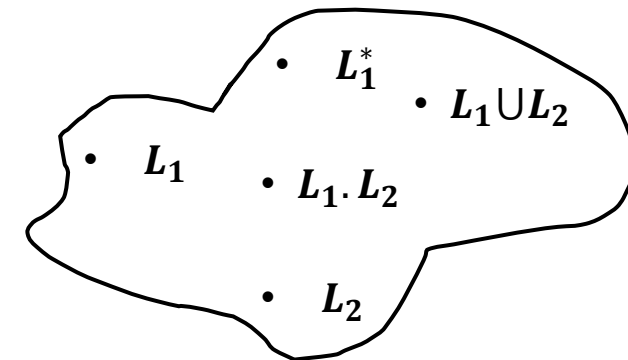
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Steps:

- Make ϵ -transitions from the final states of L_1 to the initial state of L_1 .

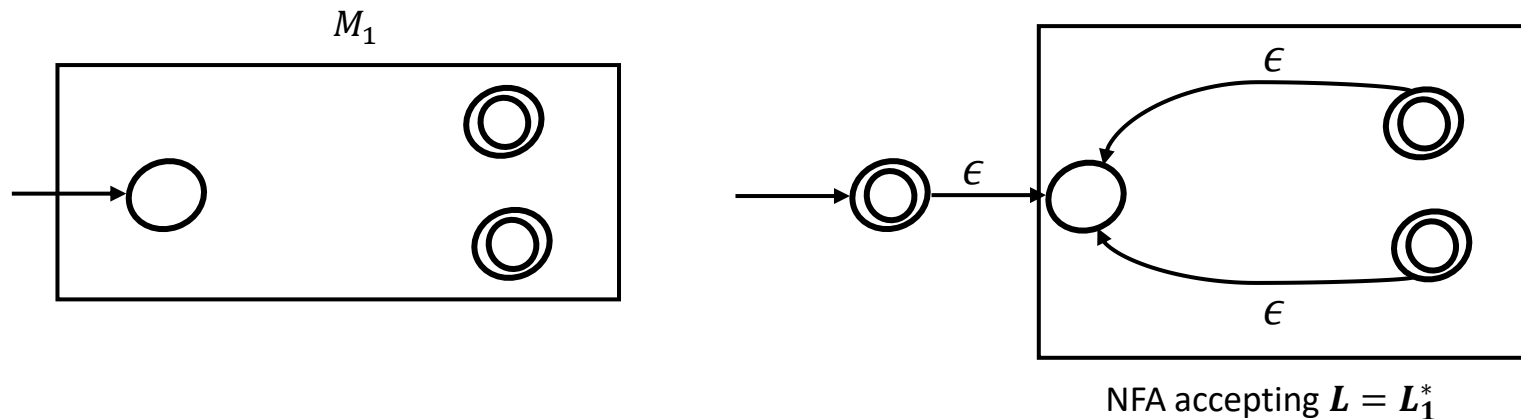


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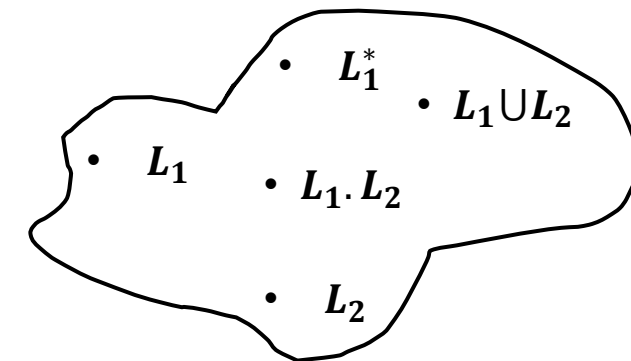
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$$L_1^* = \{x_1 x_2 \cdots x_k \mid k \geq 0 \text{ and each } x_i \in L_1\}$$

Steps:

- Make ϵ -transitions from the final states of L_1 to the initial state of L_1 .
- Make a new final state as the start state and make an ϵ -transition from this state to the previous start state of L_1 .



Set of all regular Languages

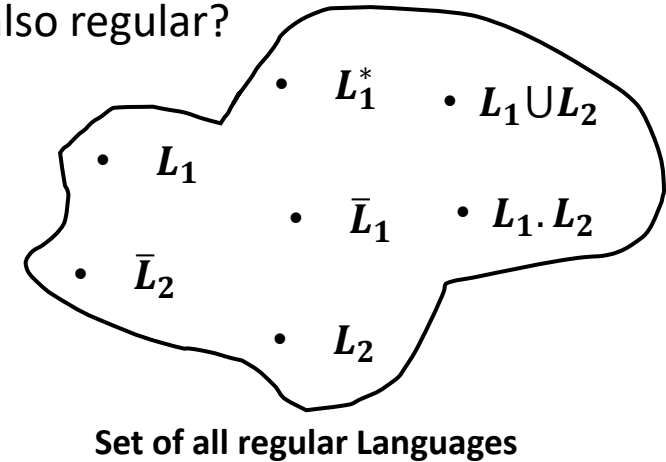
Closure of Regular Languages

Q: Is the set of all regular languages **closed under complement**? If L is regular, then is $\bar{L}$ also regular?

Proof: Given a DFA M , such that $L(M) = L$, construct the **toggled DFA** M' from M , by

- (i) changing all the non-final states of M to be the final states of M' and
- (ii) changing all the final states M to be the non-final states of M' .

$$L(M') = \bar{L}$$



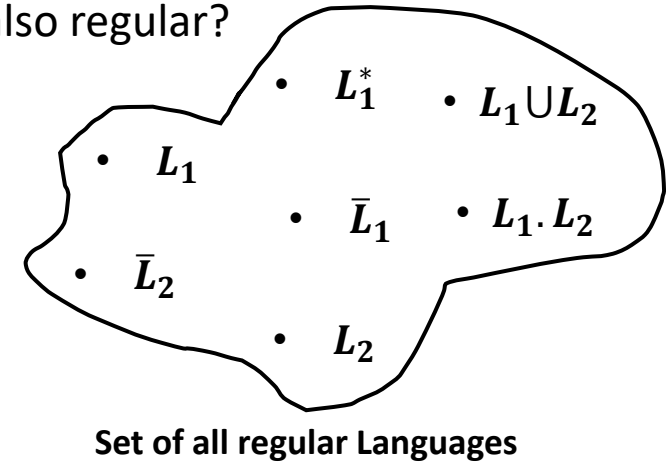
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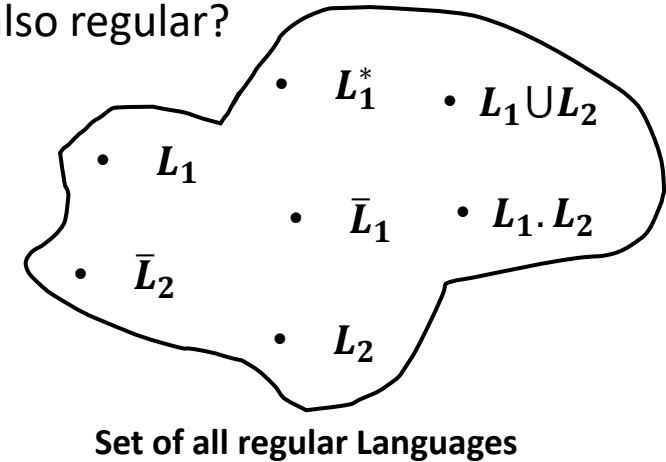
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Run 2	Accepting	

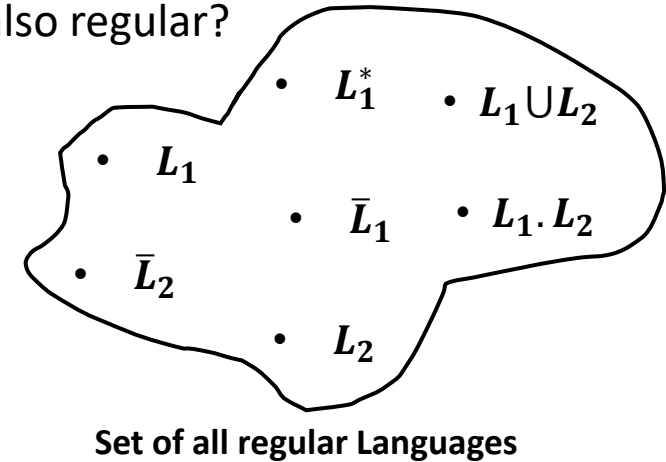
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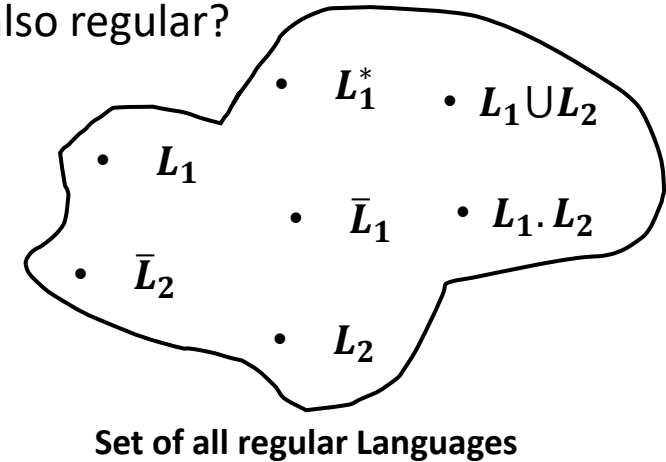
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Contradiction! So No, the **toggled NFA does not accept $\bar{L}$** .

	NFA N	Toggled NFA N'
Run 1	Rejecting	Accepting
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Closure of Regular Languages

Q: Is the set of all regular languages **closed under intersection**? If L_1 and L_2 are regular, then is $L = L_1 \cap L_2$ also regular?

Proof: We shall use the fact that regular languages are **closed** under union and complement.

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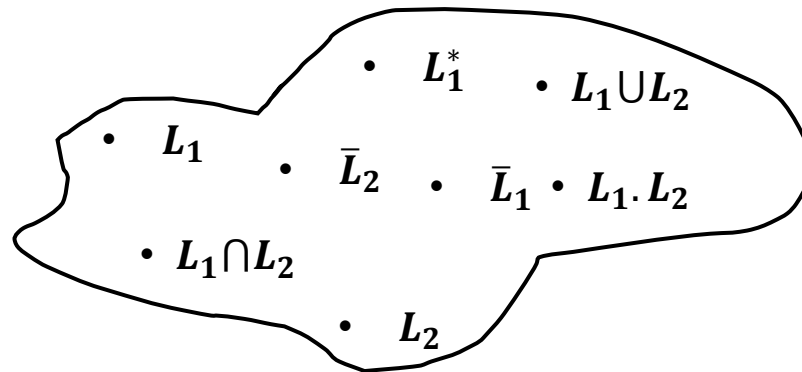
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Given a DFA for L_1 and a DFA for L_2 , we know how to construct an NFA for $\overline{L_1}, \overline{L_2}$ as well as for $L_1 \cup L_2$. Using these constructions and the aforementioned relationship, we can construct an NFA for $L = L_1 \cap L_2$



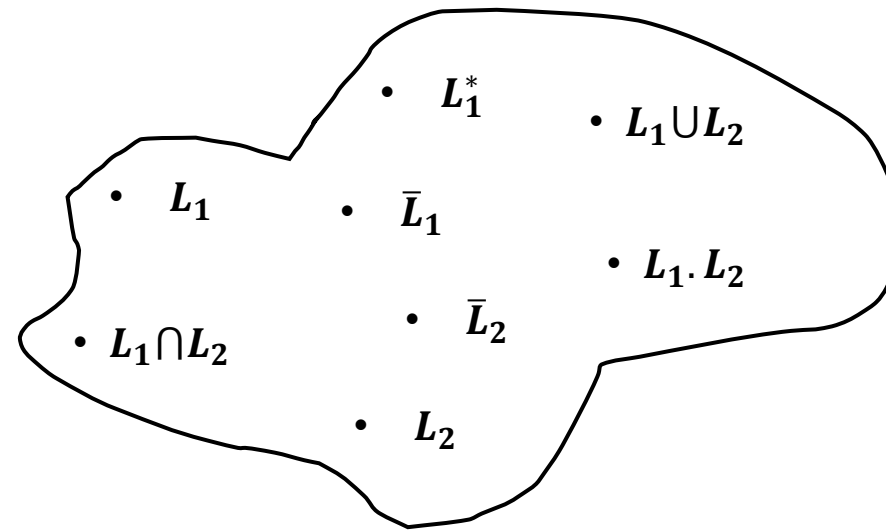
Set of all regular Languages

Closure of Regular Languages

Summary:

Regular Languages are closed under:

- **Union**
- **Intersection**
- **Star**
- **Complement**
- **Concatenation**



Set of all regular Languages

Regular Languages

If Σ is an alphabet, then

- $\Sigma^0 = \{\epsilon\}$
- $\Sigma^2 = \{a_1 a_2 | a_1 \in \Sigma, a_2 \in \Sigma\}$
- $\Sigma^k = \{a_1 a_2 \cdots a_k | a_i \in \Sigma | 1 \leq i \leq k\}$
- $\Sigma^* = \{\cup_{i \geq 0} \Sigma^i\} = \{\Sigma^0 \cup \Sigma^1 \cup \Sigma^2 \cdots\} = \{a_1 a_2 \cdots a_k | k \in \{0, 1, \cdots\} \text{ \& } a_j \in \Sigma, \forall j \in \{1, 2, \cdots, k\}\}$

A Language $L \subseteq \Sigma^*$ and $L^* = \{\cup_{i \geq 0} L^i\}$

Regular Languages

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Regular Language (alternate definition): Let Σ be an alphabet. Then the following are the regular languages over Σ :

- The empty language Φ is regular
- For each $a \in \Sigma$, $\{a\}$ is regular.
- Let L_1, L_2 be regular languages. Then $L_1 \cup L_2$, $L_1 \cdot L_2$, L_1^* are regular languages.

Regular Expressions

A regular expression describes regular languages algebraically. The algebraic formulation also provides a powerful set of tools which will be leveraged to prove

- languages are regular
- derive properties of regular languages

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Syntax for regular expressions (Recursive definition): R is said to be a regular expression if it has one of the following forms:

- Φ is a regular expression, $L(\Phi) = \Phi$
- ϵ is a regular expression, $L(\epsilon) = \{\epsilon\}$
- Any $a \in \Sigma$ is a regular expression, $L(a) = \{a\}$

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- Any $a \in \Sigma$ is a regular expression, $L(a) = \{a\}$
- $R_1 + R_2$ is a regular expression if R_1 and R_2 are regular expressions, $L(R_1 + R_2) = L(R_1) \cup L(R_2)$
- R^* is a regular expression if R is a regular expression, $L(R^*) = (L(R))^*$

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- $R_1 R_2$ is a regular expression if R_1 and R_2 are regular expressions, $L(R_1 R_2) = L(R_1) \cdot L(R_2)$
- (R) is a regular expression if R is a regular expression, $L((R)) = L(R)$

Regular Expressions

Syntax for regular expressions:

Regular Expression	Regular Language	Comment
Φ	$\{\}$	The empty set
ϵ	$\{\epsilon\}$	The set containing ϵ only
a	$\{a\}$	Any $a \in \Sigma$
$R_1 + R_2$	$L(R_1) \cup L(R_2)$	For regular expressions R_1 and R_2
$R_1 R_2$	$L(R_1) \cdot L(R_2)$	For regular expressions R_1 and R_2
R^*	$(L(R))^*$	For regular expressions R
(R)	$L(R)$	For regular expressions R

Order of precedence: $()$, $*$, $\cdot$, $+$

A language L is regular if and only if for some regular expression R , $L(R) = L$.

RE's are equivalent in power to NFAs/DFAs

Regular Expressions

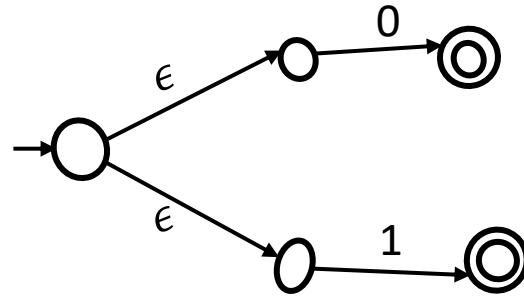
Syntax for regular expressions:

Regular Expression R	$L(R)$
01	$\{01\}$
$01 + 1$	$\{01, 1\}$
$(0 + 1)^*$	$\{\epsilon, 0, 1, 00, 01, \dots\}$
$(01 + \epsilon)1$	$\{011, 1\}$
$(0 + 1)^*01$	$\{01, 001, 101, 0001, \dots\}$
$(0 + 10)^*(\epsilon + 1)$	$\{\epsilon, 0, 10, 00, 001, 010, 0101, \dots\}$

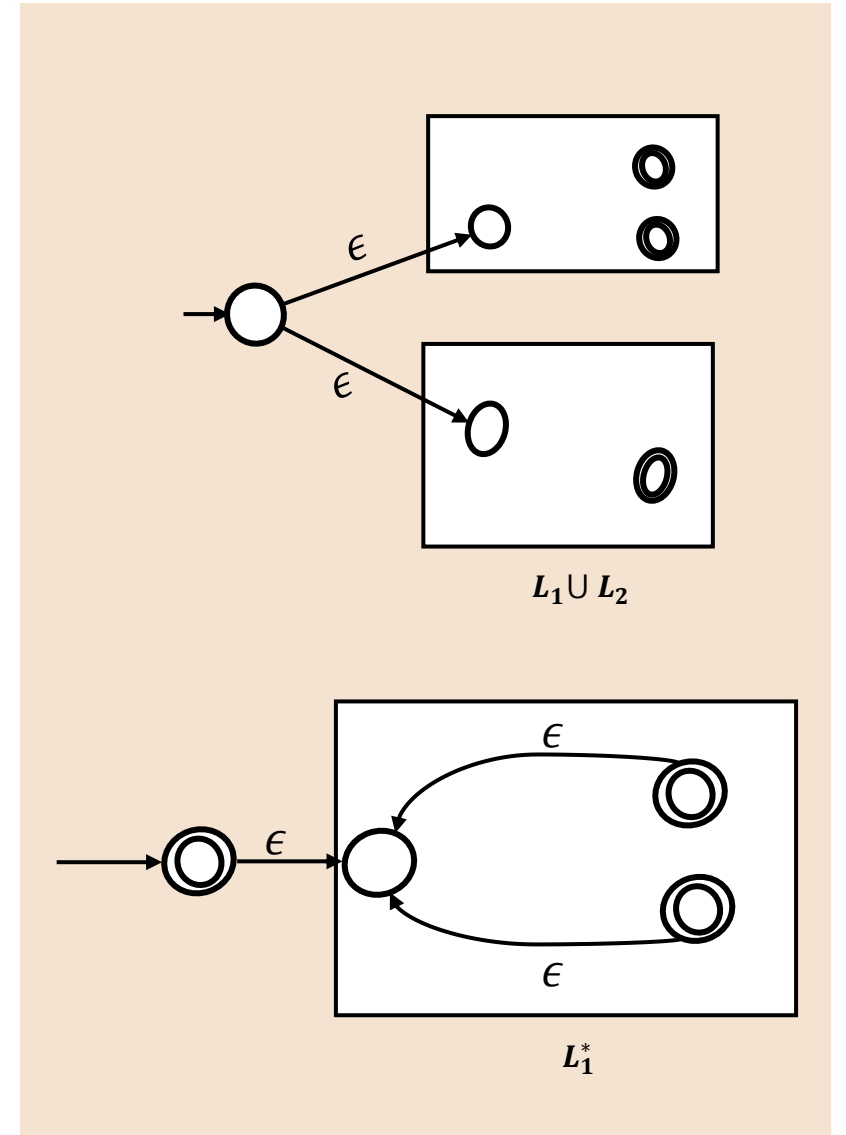
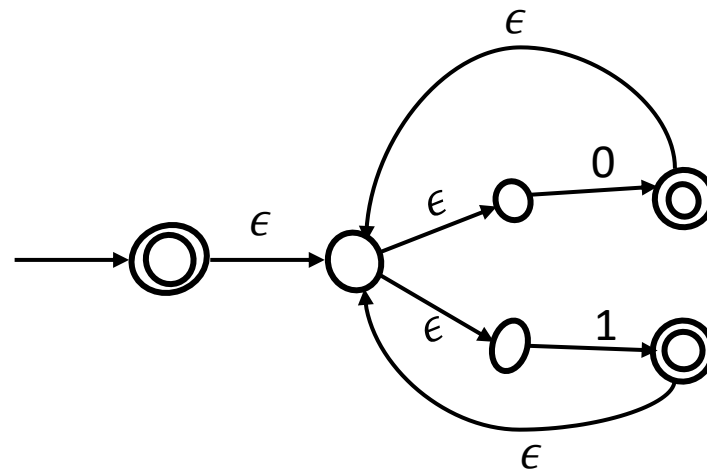
Regular Expressions

NFA for RE: $(0 + 1)^* 01$

(i) NFA for $(0 + 1)$

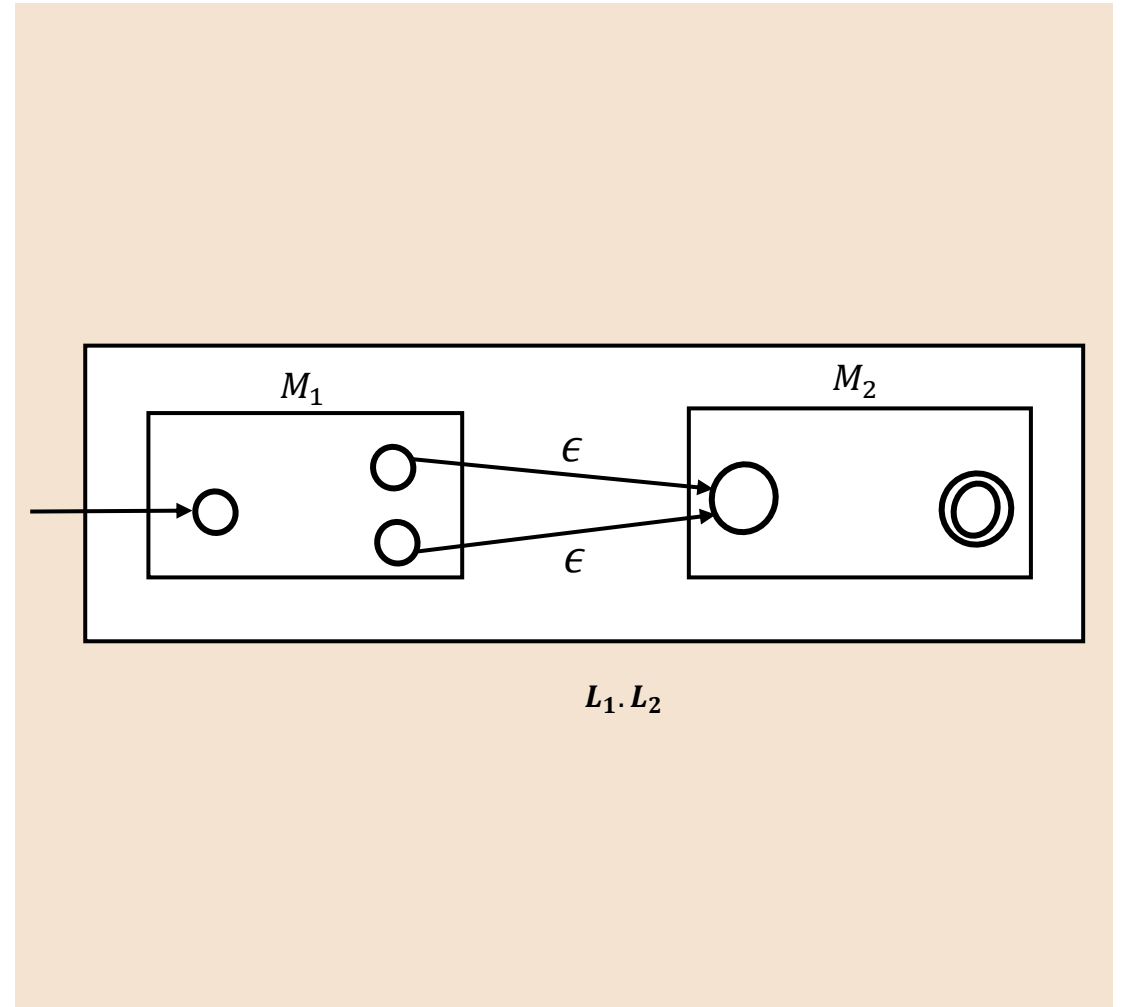
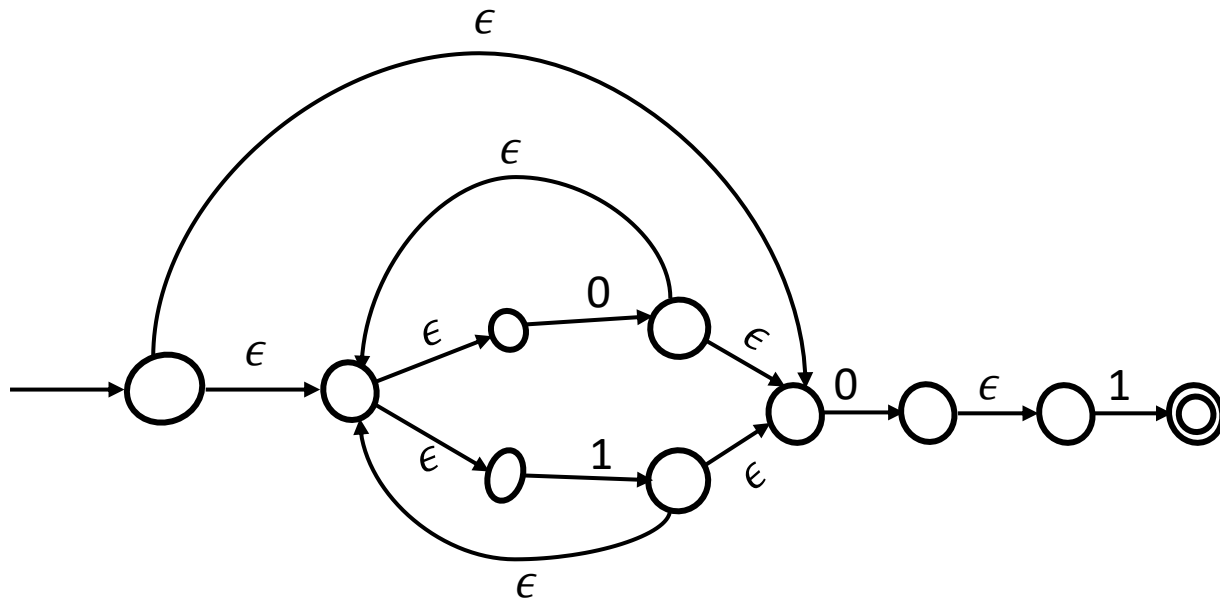


(ii) NFA for $(0 + 1)^*$



Regular Expressions

NFA for $(0 + 1)^*01$



Regular Expressions

Let $\Sigma = \{a, b\}$.

Language	Regular Expression
$\{\omega \mid \omega \text{ ends in "ab"}\}$	$(a + b)^*ab$
$\{\omega \mid \omega \text{ has a single } a \}$	b^*ab^*
$\{\omega \mid \omega \text{ has at most one } a\}$	$b^* + b^*ab^*$
$\{\omega \mid \omega \text{ is even}\}$	$((a + b)(a + b))^* = (aa + bb + ab + ba)^*$
$\{\omega \mid \omega \text{ has "ab" as a substring}\}$	$(a + b)^*ab(a + b)^*$
$\{\omega \mid \omega \text{ is a multiple of 3}\}$	$((a + b)(a + b)(a + b))^*$

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Some algebraic properties of Regular Expressions:

- $R_1 + (R_2 + R_3) = (R_1 + R_2) + R_3$
- $R_1(R_2R_3) = (R_1R_2)R_3$
- $R_1(R_2 + R_3) = R_1R_2 + R_1R_3$
- $(R_1 + R_2)R_3 = R_1R_3 + R_2R_3$
- $R_1 + R_2 = R_2 + R_1$
- $R_1^*R_1^* = R_1^*$
- $(R_1^*)^* = R_1^*$
- $R\epsilon = \epsilon R = R$
- $R\Phi = \Phi R = \Phi$
- $R + \Phi = R$
- $\epsilon + RR^* = \epsilon + R^*R = R^*$
- $(R_1 + R_2)^* = (R_1^*R_2^*)^* = (R_1^* + R_2^*)^*$

Thank You!